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7 March 1963

MILITARY SPECIFICATION

# FITTINGS, FLARELESS TUBE, FLUID CONNECTION

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. - This specification establishes the requirements for flareless tube connection fittings, nuts, and sleeves for use in fluid systems (see 6.1).

1.2 Classification. - Fittings shall be furnished in the styles designated by the applicable MS standards as specified herein or other engineering standard drawings approved by the procuring activity (see 6.1 and 6.2).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| NN-P-530 | Plywood, Flat-Panel |
| QQ-A-225/6 | Aluminum Alloy Bar, Rod, and Wire; Rolled, Drawn or Cold Finished, 2024 |
| QQ-A-225/9 | Aluminum Alloy Bar, Rod, Wire, and Special Shapes, Rolled, Drawn, or Cold Finished, 7075 |
| QQ-A-367 | Aluminum-Alloy, Forgings, Heat Treated |
| QQ-P-416 | Plating, Cadmium (Electrodeposited) |
| QQ-S-637 | Steel Bar, Carbon, Cold Finished (Standard Quality, Free Machining) |
| QQ-S-763 | Steel Bars, Shapes, and Forgings, Corrosion Resisting |
| PPP-B-566 | Boxes, Folding, Paperboard |
| PPP-B-585 | Boxes, Wood, Wirebound |
| PPP-B-591 | Boxes, Fiberboard, Wood-Cleated |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner |
| PPP-B-636 | Box, Fiberboard |

FSC 4730

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| | |
|---|---|
| PPP-B-665 | Boxes, Paperboard, Metal Stayed (Including Stay Material) |
| PPP-B-676 | Boxes, Setup |
| PPP-C-96 | Can, Metal, 28 Gage and Lighter |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-P-130 | Paper, Wrapping, Laminated and Creped |
| MIL-S-5002 | Surface Treatments and Metallic Coatings for Metal Surfaces of Weapons Systems |
| MIL-H-5606 | Hydraulic Fluid, Petroleum Base, Aircraft, Missile Ordnance |
| MIL-H-6083 | Hydraulic Fluid, Petroleum Base, for Preservation and Testing |
| MIL-H-6088 | Heat Treatment of Aluminum Alloys |
| MIL-S-6758 | Steel, Chrome-Molybdenum (4130) Bars and Reforging Stock (Aircraft Quality) |
| MIL-T-6845 | Tubing; Steel, Corrosion-Resisting (304) Aerospace Vehicle Hydraulic System, 1/8 Hard Condition |
| MIL-T-8504 | Tubing, Steel, Corrosion-Resistant (304) Aerospace Vehicle Hydraulic Systems, Annealed, Seamless and Welded |
| MIL-A-8625 | Anodic Coatings, for Aluminum and Aluminum Alloy |
| MIL-S-8879 | Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for |
| MIL-L-10547 | Liners, Case, and Sheet, Overwrap, Water-Vaporproof, or Waterproof, Flexible |
| MIL-B-13239 | Barrier Material, Waterproofed, Flexible All Temperature |
| MIL-P-17667 | Paper, Wrapping, Chemically Neutral (Non-Corrosive) |

STANDARDS

Federal

| | |
|---|---|
| FED. STD. NO. 66 | Steel: Chemical Composition and Hardenability |

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MS21902 | Union, Flareless Tube |
| MS21908 | Elbow, Bulkhead Universal, 90°, Flareless Tube |
| MS21912 | Tee, Flareless Tube with Bulkhead on Run |
| MS21914 | Cap, Pressure Seal, Flareless Tube Fitting |
| MS21921 | Nut, Sleeve Coupling, Flareless |
| MS21922 | Sleeve, Coupling Flareless |
| MS33649 | Bosses, Fluid Connection - Internal Straight Thread |

(See Supplement 1A for list of additional applicable MS Standards.)



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PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 425 . MS Fittings, Classification of Defects of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Ill. 60606.)

Society of Automotive Engineers

AMS 5643 Steel Bars, Forgings, and Rings, Corrosion Resistant, 17 Cr - 4 Ni - 4 Cu

(Copies of SAE publications may be obtained from the Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, N.Y. 10001.)

American Society for Testing and Materials

A576-67T Special Quality Hot-Rolled Carbon Steel Bars

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pennsylvania 19103.)

United States of America Standards Institute

B46.1 - 1962 Surface Texture (Surface Roughness, Waviness and Lay)

(Copies of the above publication may be obtained from the United States of America Standards Institute, 10 E. 40th St., New York, N.Y. 10016.)



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## 3. REQUIREMENTS

3.1 Qualification. - The fittings, nuts, and sleeves furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.4 and 6.3).

3.2 Materials. - Fittings, nuts, and sleeves shall be supplied from materials listed in table I, as specified on the applicable standards.

TABLE I.. Materials

| Type of part | Material | Form | Specification |
|---|---|---|---|
| Straight fittings and nuts | Aluminum Alloy | Bars, rods | QQ-A-225/6 (2024) temper T6, T62 or T851<br>QQ-A-225/9 (7075) temper T73 (Material code W) |
| Forged fittings and nuts | Aluminum Alloy | Forgings 1/ | QQ-A-367 (2014) temper T6<br>QQ-A-367 (7075) temper T73 (Material code W) |
| Nuts and straight or forged fittings | Carbon Steel | Bars, rods, plates, and forgings | 2/ QQ-S-637 (C-1137)<br>2/ QQ-S-637 (C-1141)<br>ASTM A576-67T (C-1137)<br>ASTM A576-67T (C-1141)<br>MIL-S-6758 (4130) |
| Nuts and straight or forged fittings | Corrosion-resistant steel | Bars and forgings | QQ-S-763, class 304 (Material code J)<br>QQ-S-763, class 316 (Material code K)<br>QQ-S-763, class 347 (Material code S) |
| Sleeves | Carbon Steel | Bars | 2/, 3/ QQ-S-637 (C-1213)<br>3/ ASTM A576-67T (C-1213) |
| Sleeves | Corrosion-resistant steel | Bars | AMS 5643 (17-4PH) |

1/ Use of forgings for nuts smaller than -24 size is prohibited.
2/ These alloys shall conform to Federal Standard No. 66.
3/ Nonleaded type only.



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3.2.1 Steel. - All carbon steel used in these fittings, nuts, or sleeves shall be made by the open-hearth or electric-furnace process.

3.2.2 Heat treatment (see 4.2.2). -

3.2.2.1 Aluminum alloy. - Aluminum alloy fittings and nuts shall be supplied in the final temper as shown in table II. When fitting material is purchased in other than the final temper, the desired temper is attained by aging the starting temper as shown in table II at the specified temperature for the prescribed time. Except for aging time and temperature, process control shall be in accordance with MIL-H-6088.

TABLE II. Heat treatment

| Material | Starting temper | Time (hrs.) | Temperature (degrees F) | Final temper |
|---|---|---|---|---|
| QQ-A-367 (2014) | T4 | 8 to 12 | 340 to 360 | T6 |
| QQ-A-225/6 (2024) | T4 | 12 to 14 | 370 to 380 | T6 |
| QQ-A-225/6 (2024) | T42 | 16 to 20 | 370 to 380 | T62 |
| QQ-A-225/6 (2024) | T351 | 12 to 14 | 370 to 380 | T851 |
| QQ-A/225/9 (7075) | --- | --- | --- | T73 |
| QQ-A-367 (7075) | --- | --- | --- | T73 |

3.2.2.2 Steel. - Mechanical properties of carbon steel and corrosion-resistant steel fittings, nuts, and sleeves shall be as specified in 3.2.2.2.1, 3.2.2.2.2, and 3.2.2.2.3.

3.2.2.2.1 Fittings and nuts. - Carbon steel fittings and nuts shall have a hardness of Rockwell B92 to C31. Corrosion-resistant steel fittings and nuts shall have a hardness of Rockwell B80 minimum (see 4.7.10).

3.2.2.2.2 Sleeves (carbon steel). - Carbon steel sleeves shall be heat-treated to exhibit the following metallurgical properties (see 4.7.6.1):

(a) Mean case depth of 0.0014 to 0.0020 inch. (The mean is the average thickness of the case; not the average of the extremes of the case thickness.)
(b) Case hardness of 576-754 Knoop hardness number (KHN).
(c) Core hardness of 208-245 KHN.

3.2.2.2.3 Sleeves (corrosion-resistant steel). - Corrosion-resistant steel sleeves shall be treated in such manner as to produce a minimum through-hardness of 414 KHN (see 4.7.6.2).

3.3 Design and dimensions. - The design and dimensions of fittings, nuts, and sleeves shall be in accordance with the applicable MS standard.



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3.3.1 Passages. -

3.3.1.1 Drill offset. - On straight fittings where the fluid passage is drilled from each end, the offset between the drilled holes at the meeting point of the drills shall not exceed 0.015 inch. It shall be possible to pass through the fluid passage a ball whose diameter is 0.020 less than the minimum diameter specified for the passage. This does not mean that the drilled passage may be smaller than that required by the MS standard.

3.3.1.2 On angle fittings, the cross-sectional area at the junction of the fluid passages shall be not smaller than the cross-sectional area of the smaller passage.

3.3.1.3 Wall thickness. - Except as otherwise specified on the applicable drawing, the wall thickness at any point on the fittings, nuts, or sleeves shall be not less than the thickness established by the dimensions and tolerances for the inside and outside diameters and eccentricities specified on the applicable standard.

3.3.2 Threads. - Threads shall conform to MIL-S-8879. Threads may be cut, ground, or rolled.

3.3.2.1 External threads (if rolled). - The grain flows in rolled threads shall be continuous and shall follow the general thread contour, with the maximum density at the thread root as shown on figure 1. Laps and seams, whose depths are within the limits of table III, are acceptable on the crest, and the non-pressure thread flank above the pitch diameter. Laps and seams are not acceptable on any part of the pressure thread flank, in the thread root, or on the non-pressure thread flank extending from above to below the pitch diameter. Stress cracks are unacceptable.

TABLE III. Depth of laps, seams, surface irregularities, and discontinuities in rolled threads

| Fitting size | Depth (inch) (max.) |
|---|---|
| -2 | 0.006 |
| -3 | .006 |
| -4 | .007 |
| -5 | .007 |
| -6 | .008 |
| -8 | .009 |
| -10 thru -16 | .010 |



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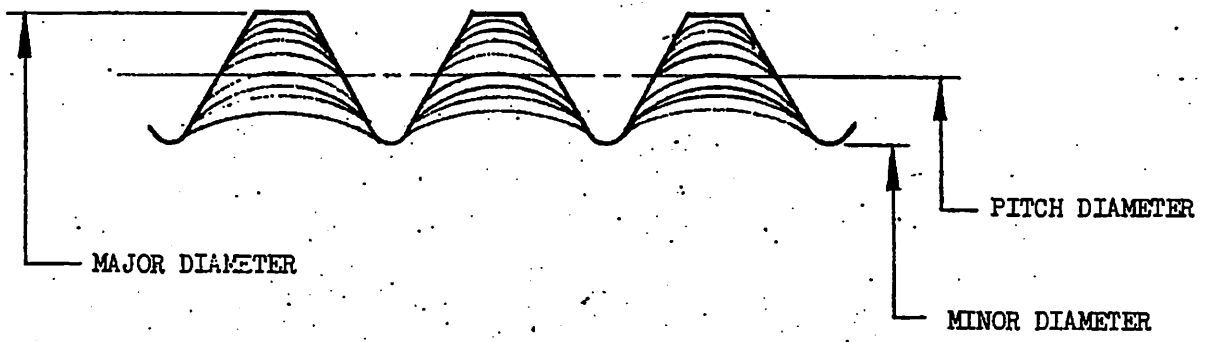



FIGURE 1. Grain flow in rolled threads



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3.4 Finish. -

3.4.1 Aluminum-alloy fittings and nuts. - Aluminum-alloy fittings and nuts (2014, 2024) shall be anodized in accordance with MIL-A-8625, type II, class 1. Aluminum-alloy fittings and nuts (7075) shall be anodized in accordance with MIL-A-8625, type II, class 2.

3.4.1.1 Surface condition. - The surface of aluminum-alloy parts, after anodizing, sealing, and surface treatment, shall be free of pits, powdery coatings, discontinuities such as scratches or breaks and shall be uniform in appearance. Racking marks shall not be on seal areas, threads, or on areas exposed to the atmosphere.

3.4.2 Carbon steel fittings. - Carbon steel fittings and nuts shall be cadmium plated in accordance with QQ-P-416, type II, class 2. Carbon steel sleeves shall be cadmium plated 0.00030 inch minimum to 0.00050 inch maximum, in accordance with QQ-P-416, type II. All carbon steel fittings, except sleeves, shall be dipped in oil conforming to MIL-H-6083.

3.4.3 Corrosion-resistant steel fittings. - Corrosion-resistant steel fittings, nuts, and sleeves shall be surface-treated in accordance with MIL-S-5002.

3.5 Identification of product. - All parts shall be identified in accordance with the instructions specified in 3.5.1, 3.5.2, 3.5.3, 3.5.4, and 3.5.5 (see 4.7.1).

3.5.1 MS standard symbol and manufacturer's trademark. - Unless otherwise specified, all fittings and nuts shall be marked with the letters "MS" and the manufacturer's name or trademark. Sleeves shall be marked with the letter "B" and the manufacturer's trademark.

3.5.2 Material identification. - Corrosion-resistant steel fittings and nuts shall be marked with the letter "J" if made from class 304, the letter "K" if made from class 316, or the letter "S" if made from class 347 corrosion-resistant steel. All 7075-T73 aluminum alloy parts shall be marked with the letter "W". The letter "D" on all other aluminum alloy parts is optional.

3.5.3 Marking for part number and size. - A numerical equivalent to the dash number indicating size is optional. All fittings and nuts larger than 3/8-inch tube size (except MS21914 fitting) shall be marked with the part number, exclusive of size. MS21914 fitting may have "MS21921" marked on the nut.

3.5.4 Size, method, and location of marking. - Marking shall be accomplished by embossing or impression stamping on the fitting or nut in a location not detrimental to the part or to its corrosion protection.



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3.5.5 Color identification. - MS fittings, nuts, and sleeves shall also be identified by the following color or finish:

Carbon steel fittings and nuts: Natural cadmium plate.
Carbon steel sleeves: No artificial dye; however, parts must be yellow in color as a result of the chromate post-plate treatment (see 3.4.2).
Corrosion-resistant steel: Natural finish.
Aluminum alloy 2014 and 2024: No artificial dye; however, parts must be yellow in color as a result of the dichromate sealer (see 3.4.1). Aluminum alloy 7075 fittings and nuts shall be dyed brown.

3.6 Performance. - The fitting, nut, and sleeve, when assembled to tubing specified in table IV, and tested in accordance with applicable procedures specified in section 4, shall be capable of the performance specified in 3.6.1, 3.6.2, 3.6.3, and 3.6.4.

3.6.1 Burst pressure. - Burst pressure shall be as shown in table IV. The fittings shall withstand burst pressure without leakage or failure (see 4.7.4).

3.6.2 Impulse. - Fittings of all materials shall be capable of 200,000 impulse cycles without leakage or failure (see 4.7.3).

3.6.3 Stress corrosion. - Aluminum-alloy fittings and nuts, when assembled to the maximum torque values specified in 4.7.2.4, shall be capable of withstanding the stress corrosion test specified in 4.7.9 for a minimum period of 100 hours without evidence of stress corrosion cracking of the nut, fitting, or sleeve.

3.6.4 Flexural strength. - At least one of the six samples of each of the assembly material combinations (tube, fitting, nut, and sleeve) shown in table IV shall withstand 10,000,000 cycles of flexure, without leakage or failure of the tube, fitting, nut, or sleeve, when tested in accordance with 4.7.7. Tube failure external to the sleeve, prior to 10,000,000 cycles, is acceptable on any of the other five samples. Leakage or failure of any fitting, nut, or sleeve on any sample prior to 10,000,000 cycles shall constitute failure.

3.6.5 Repeated assembly. - Fittings of all materials shall be capable of eight repeated assemblies, at both the minimum and maximum torque values, without leakage or indication of failure of the nut, sleeve, or fitting (see 4.7.8).

3.6.5.1 Minimum opening. - During any of the repeated assembly tests, the internal passage of neither the fitting nor the tubing shall be reduced to a value less than that shown in table V (see 4.7.8.1).



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TABLE IV. Fittings, tube material, pressures, and stresses

| Stainless-steel tube (MIL-T-6845) 1/8 hard | | | | | | | Stainless-steel tube (MIL-T-8504) annealed 1/ | | |
|---|---|---|---|---|---|---|---|---|---|
| Tube OD (in.) | Dash size | Tube wall thickness (in.) | For use with carbon steel fittings, nuts, and sleeves | | For use with aluminum-alloy fittings and nuts - carbon steel sleeves | | For use with corrosion-resistant steel fittings, nuts, and sleeves | | |
| | | | Working pressure psi 2/ | Stress level in tube for flexure test +0 % -10 psi | Working pressure psi 2/ | Stress level in tube for flexure test +0 % -10 psi | Tube wall thickness (in.) | Working pressure psi 2/ | Stress level in tube for flexure test +0 % -10 psi |
| 1/8 | - 2 | 0.012 | 3,000 | --- | 3,000 | --- | 0.020 | 3,000 | --- |
| 3/16 | - 3 | .016 | 3,000 | 35,000 | 3,000 | 15,000 | .020 | 3,000 | 20,000 |
| 1/4 | - 4 | .020 | 3,000 | ↓ | 3,000 | ↓ | .028 | 3,000 | ↓ |
| 5/16 | - 5 | .020 | 3,000 | | 3,000 | | .028 | 3,000 | |
| 3/8 | - 6 | .028 | 3,000 | | 3,000 | | .035 | 3,000 | |
| 1/2 | - 8 | .035 | 3,000 | | 3,000 | | .042 | 3,000 | |
| 5/8 | -10 | .042 | 3,000 | | 3,000 | | .058 | 3,000 | |
| 3/4 | -12 | .058 | 3,000 | | 3,000 | | .065 | 3,000 | |
| 1 | -16 | .065 | 3,000 | | 1,500 | | .083 | 3,000 | |
| 1-1/4 | -20 | .049 | 1,500 | | 1,500 | | .065 | 1,500 | |
| 1-1/2 | -24 | .058 | 1,500 | | 1,500 | | .065 | 1,500 | |
| 2 | -32 | .065 | 1,500 | | 1,500 | | .095 | 1,500 | |

1/ Qualification of one alloy (class 304, 316, or 347) constitutes qualification of the other alloys.

2/ Proof pressure is twice working pressure; burst pressure is four time working pressure.



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TABLE V. Minimum opening in inches when using tubing specified in table IV for the repeated assembly and tube cut tests

| Tube size | Specification MIL-T-6845 | Specification MIL-T-8504 |
|---|---|---|
| 2 | 0.070 | 0.060 |
| 3 | .110 | .115 |
| 4 | .165 | .155 |
| 5 | .225 | .225 |
| 6 | .290 | .270 |
| 8 | .400 | .380 |
| 10 | .485 | .475 |
| 12 | .610 | .590 |
| 16 | .840 | .800 |
| 20 | 1.080 | 1.080 |
| 24 | 1.330 | 1.330 |
| 32 | 1.780 | 1.780 |

3.6.6 Tube cut. - After assembly in accordance with 4.7.5, disassembly and inspection of the sleeve shall disclose no longitudinal or circumferential cracks on the sleeve. Circumferential crazing marks are permissible. The tube collapse shall not exceed the minimum openings specified in table V. After driving back the sleeve, the exposed ring cut on the tube shall encircle 100 percent of the tube periphery.

3.7 Workmanship. - Machined surfaces of fittings, nuts, and sleeves, shall be free from burrs, longitudinal or spiral tool marks. Unless a finer finish is specified on applicable drawings, sealing surfaces shall be smooth, except that annular tool marks up to 100 microinches roughness height rating (RHR), as defined in USA B46.1 - 1962, will be acceptable. All other machined surfaces shall not exceed 125 RHR. Unmachined surfaces, such as forging surfaces and bar stock flats, shall be of uniform quality and condition, free from blisters, fins, folds, seams, laps, cracks, segregations, spongy areas, or other defects which would adversely affect their serviceability, and, except for forging parting lines, shall not exceed 250 RHR. Surface defects may be explored and if they can be removed so that they do not appear on re-etching and the required section thickness can be maintained, they shall not be cause for rejection. The 35-degree sloped bearing surface, the internal surface from the cutting edge to a point 1/16 inch toward the tail end, and the external sealing surface, from the tip to within 1/16 inch of the shoulder, of all sleeves shall be finished to 63 RHR, as defined in USA B46.1 - 1962 (see 4.7.1).



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. - A lot shall consist of all parts of a given part number made from the same batch of material and processed at the same time.

4.2.1 Record maintainance. - The supplier shall maintain a record of inspections applied to each lot.

4.2.2 Material certification. - Records of chemical composition analysis, mechanical property tests showing conformance to the applicable material specifications, and data showing conformance to the test requirements specified in MIL-H-6088, shall be made available to the procuring activity for each lot of fittings.

4.3 Classification of inspections. - The inspection and testing of flareless fittings, nuts, and sleeves shall be classified as follows:

(a) Qualification inspection (4.4)
(b) Quality conformance inspection (4.5)

4.4 Qualification inspection. -

4.4.1 Sampling instructions. - Qualification inspection samples shall consist of the parts specified in table VI, for each size and material for which qualification is desired.

4.4.2 Tests. - Qualification inspection shall consist of the following requirements and tests, as applicable, and as specified in sections 3 and 4:

| | | |
|---|---|---|
| (a) | All quality conformance inspection | (4.5.2) |
| (b) | Burst pressure | (3.6.1, 4.7.4) |
| (c) | Repeated assembly | (3.6.5, 4.7.8) |
| (d) | Stress corrosion | (3.6.3, 4.7.9) |
| (e) | Flexural strength | (3.6.4, 4.7.7) |
| (f) | Impulse | (3.6.2, 4.7.3) |
| (g) | Grain flow | (3.3.2.1, 4.7.11) |
| (h) | Tube cut | (3.6.6, 4.7.5) |



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TABLE VI. Qualification test samples 1/

| Quantity (each) 2/ | Part No. | Material | Applicable test |
|---|---|---|---|
| 3 | MS21908<br>MS21912 | 5/ | Burst pressure (see 3.6.1 and 4.7.4) |
| 3 | MS21908<br>MS21912 | 5/ | Impulse (see 3.6.2 and 4.7.3) |
| 3 | MS21912<br>MS21902 | 3/ | Stress corrosion (see 3.6.3 and 4.7.9) |
| 6 | MS21902 | 5/ 8/ | Flexural strength (see 3.6.4 and 4.7.7) |
| 6 | MS21902 | 5/ | Repeated assembly (see 3.6.5 and 4.7.8) |
| 6 | MS21922 | 5/ | Tube cut (see 3.6.6 and 4.7.5) |
| 6 | MS21902 | 4/ 6/ | Grain flow (rolled threads) (see 3.3.2.1 and 4.7.11) |
| 3 | MS21922 | 6/ | Metallurgical properties (see 3.2.2.2.2, 3.2.2.2.3, and 4.7.6) |
| 7/ | 7/ | | Quality conformance tests (a), (b), (c), and (f), of 4.5.2 (see 2/). |

1/ All fittings specified shall be assembled with applicable MS21921 nuts, MS21922 sleeves and tubing specified in table IV.
2/ In each size and material for which qualification is desired.
3/ Aluminum-alloy fittings and nuts only (2014, 2024, 7075).
4/ Carbon steel and all corrosion-resistant steels in sizes -2 to -16, inclusive.
5/ All materials as applicable (2014, 2024, and 7075 aluminum alloys) (class 316, 304, 347 corrosion-resistant steel (CRES) (any allowable carbon steel) qualification tests report must identify material used.
6/ This test specimen may be obtained from the tube cut test (4.8.5), after its completion.
7/ All samples subjected to qualification inspection tests shall be selected from a lot previously subjected to the specified quality conformance inspection ( (a), (b), (c), and (f) of 4.5.2). Sleeves, however, shall not have been subjected to test (f).
8/ For fittings of all materials in size -2, the flexural strength test shall not be required.



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4.5 Quality conformance inspection. -

4.5.1 Sampling. -

4.5.1.1 Nondestructive tests. - Sampling for material, threads, finish, dimensions, marking, surface defects, and workmanship shall be at random in accordance with MIL-STD-105 at an Acceptable Quality Level (AQL) of 4.0 percent unless otherwise specified in ANA Bulletin No. 425.

4.5.1.2 Destructive tests. - Sampling for tests specified in (d), (e), (f), (g), and (h) of 4.5.2 shall be performed in accordance with MIL-STD-105, Inspection level S-1, acceptance number zero.

4.5.2 Inspection. - Each individual lot of fittings, nuts, and sleeves shall conform to the following requirements and be subjected to the following examination and tests, as applicable, and as specified in sections 3 and 4:

| | | |
|---|---|---|
| (a) | Examination of product | (4.7.1) |
| (b) | Chemical composition | (3.2, 4.2.2) |
| (c) | Physical properties | (3.2.2.1, 4.2.2) |
| (d) | Metallurgical properties (sleeve only) | (3.2.2.2.2, 3.2.2.2.3, and 4.7.6) |
| (e) | Tube cut (sleeves only) | (3.6.6, 4.7.5) |
| (f) | Hardness | (3.2.2.2, 4.7.10) |
| (g) | Burst pressure | (3.6.1, 4.7.4) |
| (h) | Grain flow | (3.3.2.1, 4.7.11) |

4.6 Rejection and retest. - Rejected lots shall be resubmitted for retest and acceptance in accordance with MIL-STD-105. Parts subjected to nondestructive tests and failing to conform to the requirements of these tests shall be rejected. Parts subjected to destructive tests shall be discarded.

4.7 Inspection methods. -

4.7.1 Examination of product. - Each lot of fittings will be examined to determine conformance with this specification and the applicable standard with respect to material, dimensions, threads, wall thickness, surface defects, finish, marking, and workmanship (see 3.2, 3.3, 3.4, 3.5, and 3.7).

4.7.2 General testing practice for tests. -

4.7.2.1 Thread lubricant. - No thread lubricant shall be used other than hydraulic fluid conforming to MIL-H-5606 or MIL-H-6083, unless specifically approved by the procuring activity.

4.7.2.2 Tube preparation. - Tubes shall be cut square within 1/2 degree and all burrs removed from inside and outside of the tube ends. The break or chamfer on either the outside diameter or inside diameter shall not exceed 25 percent of the thickness of the tube wall.



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4.7.2.3 Presetting of sleeves. - Sleeves shall be preset on the tubing before assembly using a presetting machine, or with the fitting, by tightening the nut to the number of turns specified in table VII, past the point of ring grip.

TABLE VII. Preset - tube tightening limits

| Annealed stainless steel tubing | | | 1/8 hard stainless steel tubing | |
|---|---|---|---|---|
| Tube OD (in.) | Wall thickness (in.) | Number of turns 1/ | Wall thickness (in.) | Number of turns 1/ |
| 1/8 | 0.020 | 1-1/6 | 0.012 | 1-1/6 |
| 3/16 | .020 | 1-1/6 | .016 | 1-1/6 |
| 1/4 | .028 | 1-1/6 | .020 | 1-1/6 |
| 5/16 | .028 | 1-1/6 | .020 | 1-1/6 |
| 3/8 | .035 | 1-1/6 | .028 | 1-1/6 |
| 1/2 | .042 | 1 | .035 | 1-1/6 |
| 5/8 | .058 | 1 | .042 | 1-1/6 |
| 3/4 | .065 | 1 | .058 | 1-1/6 |
| 1 | .083 | 5/6 | .065 | 1 |
| 1-1/4 | .065 | 5/6 | .049 | 1 |
| 1-1/2 | .065 | 5/6 | .058 | 1 |
| 2 | .095 | 5/6 | .065 | 1 |

1/ Number of turns to preset tubing (without mandrels).

4.7.2.4 Resetting of sleeves. - After presetting the sleeves as specified in 4.7.2.3, the nut shall be completely disengaged and then tightened to the minimum or maximum torque values, as applicable, in accordance with table VIII.

TABLE VIII. Torque values

| Size | Torque (inch-pounds) | |
|---|---|---|
| | Minimum | Maximum |
| -2 | 75 | 85 |
| -3 | 95 | 105 |
| -4 | 135 | 145 |
| -5 | 170 | 190 |
| -6 | 215 | 245 |
| -8 | 470 | 510 |
| -10 | 620 | 680 |
| -12 | 855 | 945 |
| -16 | 1140 | 1260 |
| -20 | 1520 | 1680 |
| -24 | 1900 | 2100 |
| -32 | 2660 | 2940 |



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4.7.2.5 Tube material and working pressure. - Aluminum-alloy and carbon-steel fittings, shall be tested with carbon steel sleeves and MIL-T-6845 1/8-hard stainless steel tubing. Stainless-steel fittings and sleeves shall be tested with MIL-T-8504 annealed stainless-steel tubing. Working pressures shall be as specified in table IV.

4.7.2.6 Proof pressure. - Proof pressure shall be twice the working pressure specified in table IV.

4.7.2.7 Burst pressure. - Burst pressure shall be four times the working pressure specified in table IV.

4.7.2.8 Test fluid. - Fluid conforming to MIL-H-5606 or MIL-H-6083, shall be used in all tests.

4.7.3 Impulse. - Three samples each of MS21908 elbows and MS21912 tees shall be mounted on tubing of the type designated in table IV for the fitting materials under test. The tube assembly shall be joined with minimum torque values and subjected to the following tests.

4.7.3.1 The assembly shall first be tested at proof pressure for 5 minutes. All impulse testing shall be performed at the rate of 65 ±5 cycles per minute (cpm). Each impulse cycle shall constitute a rise from 0 pounds per square inch (psi) to surge pressure and drop to pressure approaching zero. Hydraulic fluid shall be used as the testing media. The peak surge pressure shall be within 1.43 to 1.57 times the working pressure, as shown by an oscillograph or other electronic measuring device. Fittings shall complete 200,000 impulse cycles, after which they shall again be tested at proof pressure for 5 minutes (see figure 2).

4.7.4 Burst pressure. - For qualification, MS21908 elbows and MS21912 tees, shall be used in this test. Fittings shall be assembled so that at least 8 inches of free tubing extend between the two end fittings. One-half the samples shall be assembled with minimum torque and the remaining samples assembled with maximum torque. Pressure shall be raised to the proof pressure for 5 minutes. Pressure shall then be increased at a rate of 20,000 ±5,000 psi per minute to the burst pressure, as specified in 4.7.2.7, or until burst or leakage occurs; no burst or leakage shall occur at a pressure less than the specified burst pressure. (See table IV.) For quality conformance tests, only fittings of part number being furnished need to be tested, in which case the tube, nut, and sleeve may be omitted and a suitable adapter substituted.

4.7.5 Tube cut. - Carbon steel sleeves shall be assembled to 1/8-hard corrosion-resistant steel tubing conforming to MIL-T-6845, with either aluminum-alloy or carbon steel fittings and nuts. Corrosion-resistant steel sleeves, fittings, and nuts shall be assembled to annealed corrosion-resistant steel tubing conforming to MIL-T-8504. A maximum of one-half of the sleeve samples may be assembled by machine preset methods. Remaining samples shall be assembled by the



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nd preset method. One-half of the required assemblies shall be tightened to minimum torque values and one-half to maximum torque values. After complete assembly, one-half of the maximum torque samples and one-half of the minimum torque samples shall be subjected to the burst pressure test (4.7.4). The remaining assemblies shall be subjected to the proof pressure test (4.7.2.6), the nuts shall then be removed, the tubing withdrawn, and the assembly examined for tube collapse, longitudinal or circumferential cracks on the sleeves. The sleeve shall be carefully driven back to expose the tube ring cut for examination.

4.7.6 Metallurgical properties (sleeves only). - Samples as required shall be subjected to the tests specified in 4.7.6.1 and 4.7.6.2, as applicable.

4.7.6.1 Carbon steel sleeves, mounted as indicated on figure 3, shall exhibit the metallurgical properties specified in 3.2.2.2.2, and as determined by the following methods:

(a) Case depth: The case depth determination shall consist of microscopic measurement of the case depth on a polished and etched cross section of the sleeve. A minimum of five measurements shall be taken at different points on the sleeve with at least two readings on the inner case (ID) and two on the outer case (OD). The cross section shall be taken perpendicular to the axis of the sleeve 0.03 to 0.06 inch back of the cutting edge, as illustrated on figure 3.

(b) Case hardness: The case hardness shall be measured at a point 0.0007 ±0.0002 inch in depth below the surface of the sleeve material and within the same area defined in (a) above, as illustrated on figure 3. The measurement shall be made with a Tukon hardness tester, using a Knoop indenter and a 100-gram load. A minimum of four readings shall be taken, with two on the inner case (ID) and two on the outer case (OD).

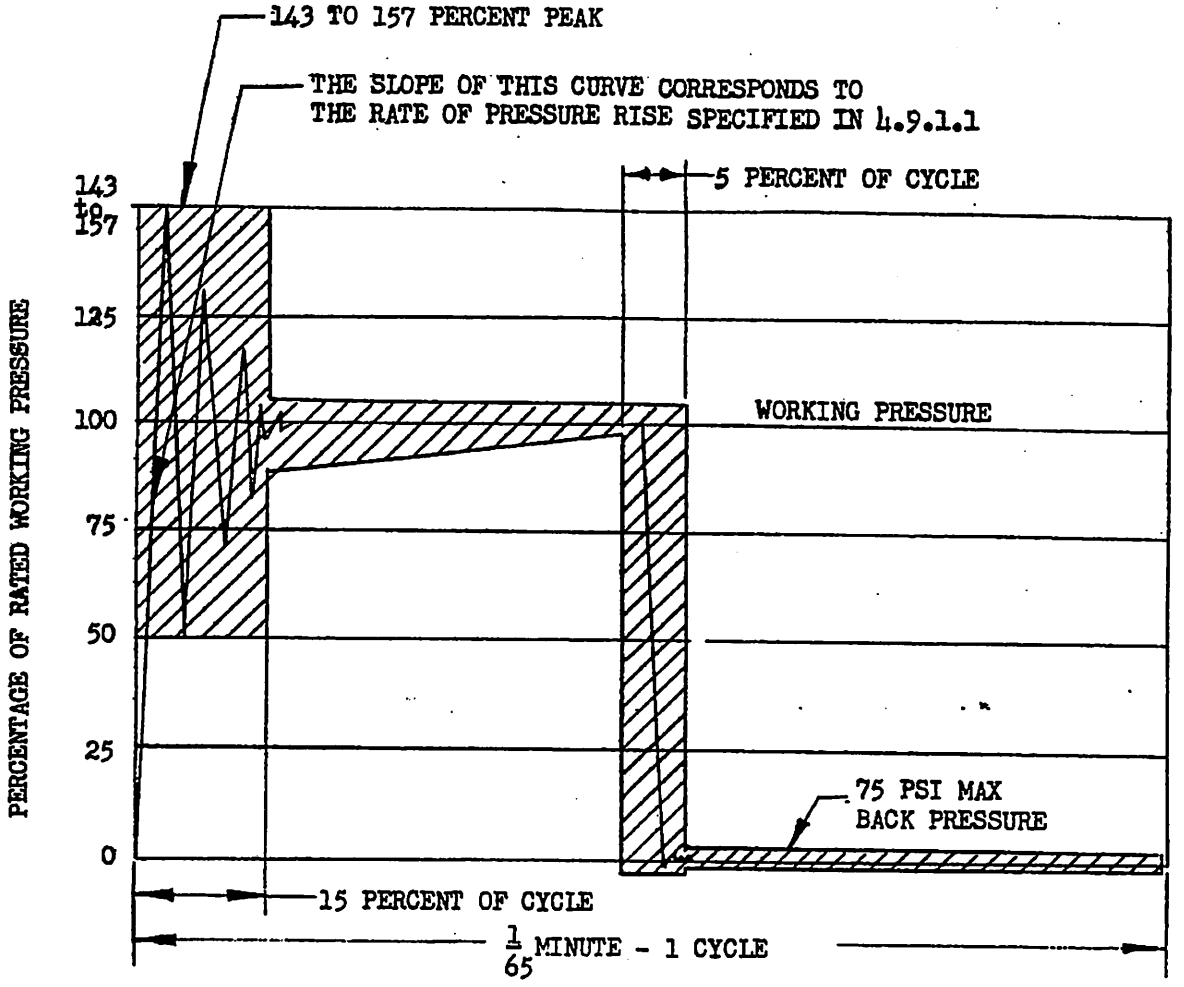
(c) Core hardness: Core hardness shall be measured at a minimum of three places on the same samples used for case hardness, and with the same equipment, except that the load shall be 500 grams.

4.7.6.2 Hardness of corrosion-resistant steel sleeves shall be measured, using samples mounted as shown on figure 3, using a load of 500 grams.

4.7.7 Flexural strength. - The flexure test may be performed in one of several ways. Two methods are shown on figures 4 and 5. Any method may be used, provided that the stress level indicated in table IV is imposed and measured at a point adjacent to a MS21902 union and MS21922 sleeve under test. The stress level measured shall include tensile bending stress plus the axial tension due to pressure. Measurement of the stress shall be obtained by placing two strain gages, 180 degrees apart, on the periphery of the tube in the plane of maximum stress. The center of the strain gage shall be placed 3/16 ±1/64 inch from the tail of the sleeve. During the test, a constant pressure equal to the working pressure, noted in table IV, shall be imposed. Frequency of flexing may be any rate from 30 to 500 cycles per second, depending upon method selected. The specimens shall be assembled with maximum torque values. Duration of test shall be 10,000,000 cycles or until prior failure or leakage occurs.



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THE CURVE SHOWN ABOVE IS THE APPROXIMATE PRESSURE-TIME CYCLE DETERMINED TO BE OF PROPER SEVERITY FOR IMPULSE TESTING OF FLARELESS FITTINGS. WHILE IT IS MANDATORY ONLY THAT PRESSURE PEAK RISES TO 143 TO 157 PERCENT OF THE WORKING PRESSURE AT SOME POINT PRIOR TO LEVELING OFF AT RATED PRESSURE, IT IS CONSIDERED HIGHLY DESIRABLE THAT THE PRESSURE-TIME CURVE BE CONFINED TO THE SHADED AREA INDICATED. ONE VERY DESIRABLE BENEFIT TO BE GAINED IN THIS MANNER IS THAT RESULTS OF TESTS PERFORMED ON DIFFERENT TEST MACHINES WILL BE MORE NEARLY COMPARABLE.

FIGURE 2. Dynamic pressure impulses



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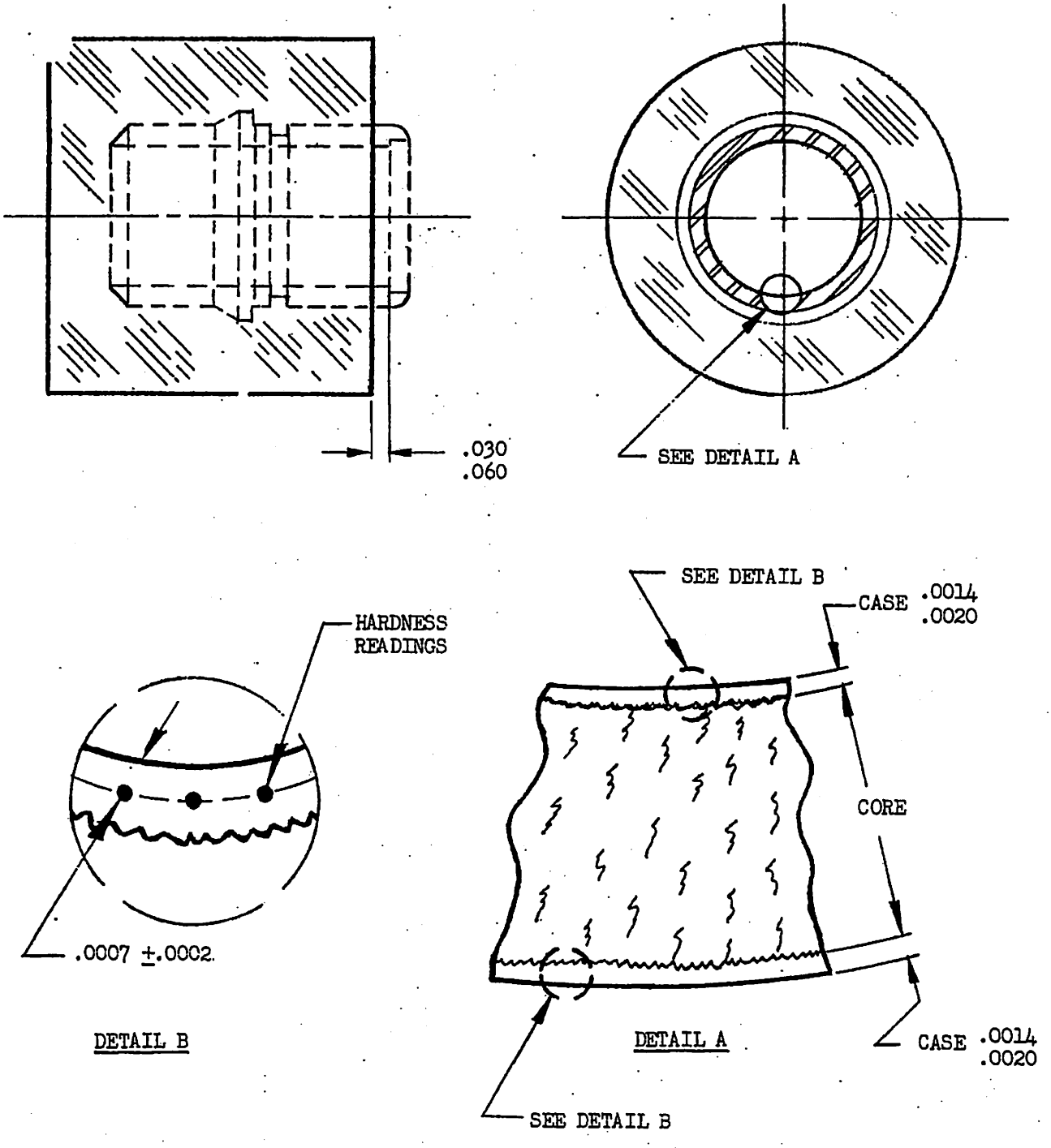



FIGURE 3. Case and core hardness



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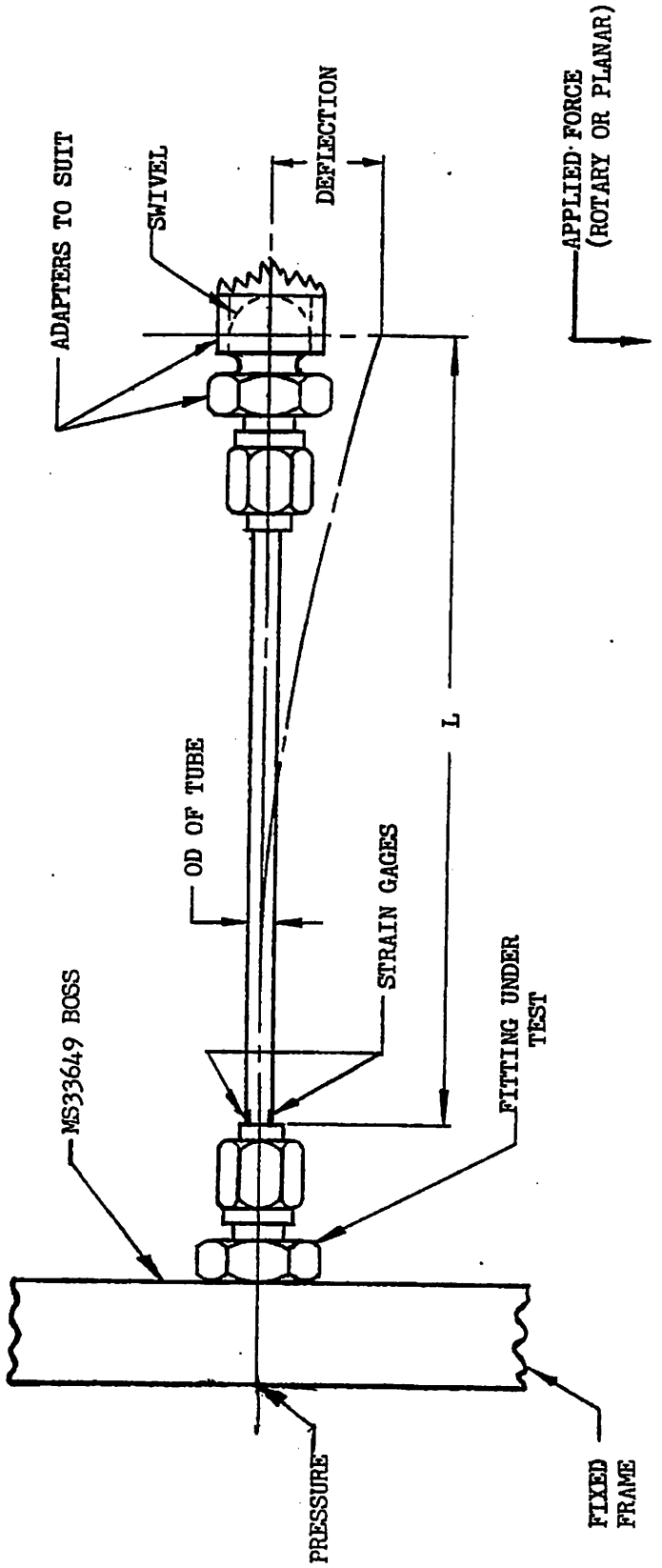



FIGURE 4. Cantilever beam method



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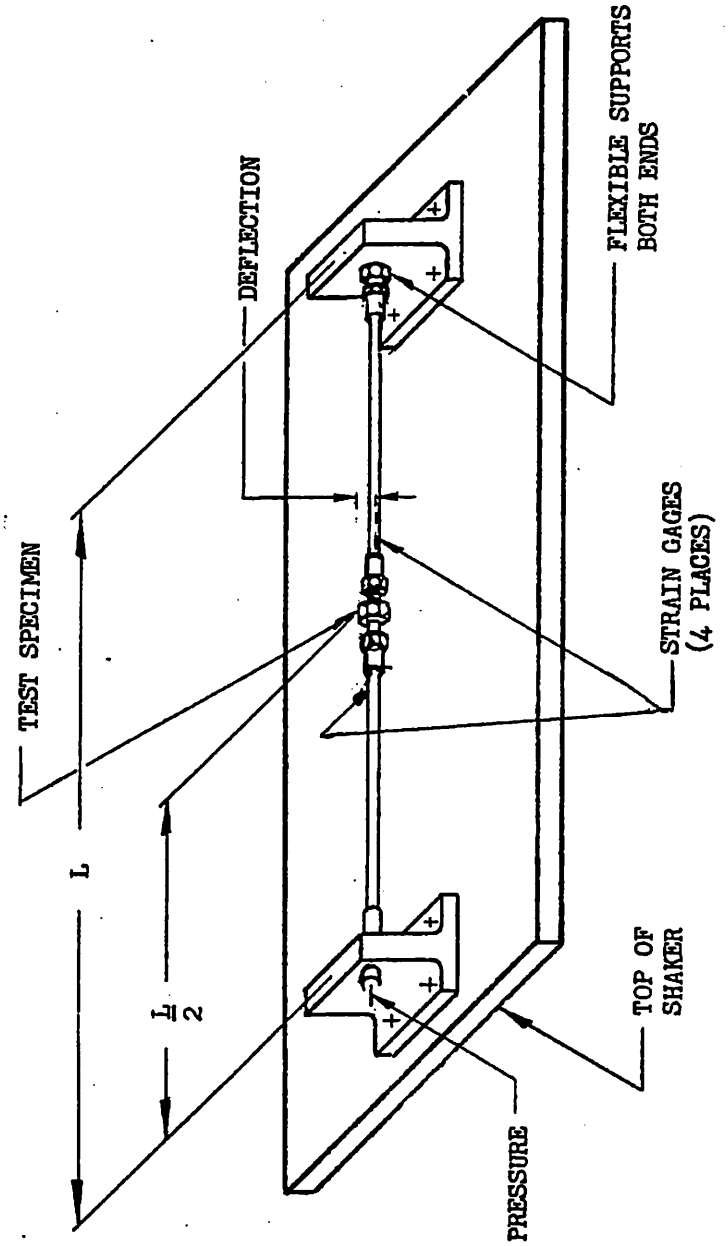



FIGURE 5. Simple beam method



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4.7.7.1 Cantilever beam method. - Figure 4 shows one method, wherein the flexural requirements are obtained by imposing a concentrated load on the free swivelling end of the tube assembly. The opposite end (test fitting) is rigidly supported in a fixture capable of varying the tube length to obtain the desired stress level in the tube. The motion of the free end may be either rotary or planar. Tube length "L" shall be determined by actual stress measured. Tentative length may be calculated, however, from the following formula:

$$L = \sqrt{\frac{3\ DEC}{S}}$$

in which L = length in inches from ℄ strain gage to point of deflection
D = Displacement in inches
E = modulus of elasticity (30,000,000 for corrosion-resistant steel)
C = one-half tube outside diameter (inches)
S = stress in tube (see table IV) (at strain gage).

The above formula was derived by combining the following formulae:

$$S = \frac{MC}{I} = \frac{FLC}{I}$$

$$D = 1/3\ \frac{FL^3}{EI}$$

4.7.7.2 Simple beam method. - Figure 5 shows another method, wherein the test arrangement is essentially a simple beam, with a concentrated load (test fitting weight times acceleration) at the center mounted on a "shaker". Both supports should provide free swivelling action. In order to eliminate side loads on the shaker table, it is recommended that the static pressure be applied to both ends of the test specimen. Total length of the test specimen "L" shall be determined by actual stress measured. Tentative length "L" may be calculated, however, from the following formula:

$$L = \sqrt{\frac{12\ DEC}{S}}$$

in which L = total length in inches
D = displacement at center in inches
E = modulus of elasticity (30,000,000 for corrosion-resistant steel)
C = one-half tube outside diameter (inches)
S = stress in tube (see table IV) (at strain gage).

Note: Displacement (D) is a function of weight of tube union times acceleration imposed. Acceleration should be varied to obtain desired displacement (stress).



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4.7.8 Repeated assembly. -

4.7.8.1 Minimum tightening. - The tube and fitting assembly shall be assembled and disassembled eight successive times, using minimum torque values specified in table VIII. Each repeated assembly operation shall include the assembly and complete removal of the tube from the fitting body and reassembly 90 degrees out of previous tube-to-fitting phase relationship. (Sleeve is not removed from tube.) After each third and final tightening operation, the assembly shall be subjected to the proof pressure for 5 minutes. The tubing shall be examined for tube collapse (see 3.6.5.1) after the seventh repeated assembly. After the eighth tightening operation, the assembly shall be subjected to the burst pressure test (see 3.6.1 and 4.7.4).

4.7.8.2 Maximum tightening. - The test specified in 4.7.8.1 shall be performed, using new assemblies, with the maximum torque values specified in table VIII.

4.7.9 Stress corrosion. - A 12-inch length of 1/8-hard corrosion-resistant steel tube, conforming to MIL-T-6845, shall be assembled with a carbon steel sleeve, aluminum-alloy nut and fitting, using the maximum torque values. The assembly shall then be pressurized to the operating pressure, which shall be maintained for the duration of the test. The assembly shall then be immersed, so that it is covered a minimum of 1 inch, in a sodium chloride solution maintained at 6.0 percent NaCl concentration at a temperature of 190° ±10° F for 5 minutes. The assembly shall be removed and dried in air, at 140° F, for 25 minutes. The immersion and drying cycle shall be repeated for a total period of 100 hours The nut, fitting, and sleeve shall be disassembled and carefully examined with a 10-power magnifying glass for evidence of cracking. The parts shall then be reassembled and subjected to the burst pressure test specified in 4.7.4.

4.7.10 Hardness. - Hardness readings shall be taken on a smooth flat surface of any unthreaded portion of the fitting or nut. Carbon steel fittings and nuts shall have the plating removed before taking hardness readings (see 3.2.2.2.1). Hardess of sleeves shall be taken in the location specified in 4.7.6.1 and 4.7.6.2.

4.7.11 Grain flow in threads. - Grain flow in rolled threads shall be determined by macroexamination. Specimens shall be taken from the finished part by sectioning on a longitudinal plane across the threaded area. The specimen shall be etched in an aqueous solution containing 50 percent (by volume) of commercial hydrochloric acid at 160° to 180° F for sufficient time to reveal the macrostructure adequately.

4.8 Packaging, packing, and marking. - Preparation for delivery shall be examined for conformance to section 5, and 4.8.1.

4.8.1 Sampling. - Sampling shall be performed in accordance with MIL-STD-105, Inspection level S-3 and acceptable quality level (AQL) of 4.0 percent defective. Classification of defects shall be as shown in table IX.



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TABLE IX. Classification of defects for preparation for delivery

| Item | Defects |
|---|---|
| Exterior and interior markings | Missing, incorrect, incomplete, illegible, of improper size, location, sequence or method of application |
| Materials | Any nonconforming component; component missing, damaged, or otherwise defective |
| Workmanship | Inadequate assembly of components |
| Exterior and interior weight or content | Number per container is more or less than stipulated; gross or net weight exceeds the requirement |

5. PREPARATION FOR DELIVERY

5.1 Preservation and cleaning.

5.1.1 Preservation. - No preservative compound shall be applied to the fittings, except that cadmium-plated steel parts shall be dipped in oil as specified in 3.4.2.

5.1.2 Cleaning. - Before packaging, all parts shall be free from grease, oil, dirt, or any other foreign matter, except as noted in 5.1.1.

5.2 Packaging. - Packaging shall be level A or level C, as specified (see 6.2).

5.2.1 Level A. -

5.2.1.1 Fitting unit containers. - All fittings except nuts, sleeves, and caps shall be unit packaged in accordance with Method IC-1 of MIL-P-116, except as modified herein. The quantity per unit package shall be one. After bagging, fittings greater than 3-1/2 inches in any dimension shall be individually overpacked in cartons conforming to PPP-B-636, PPP-B-566, PPP-B-676, or PPP-B-665. If it is necessary to wrap the fittings prior to inserting in bags to prevent damage, fittings shall be wrapped with barrier materials conforming to MIL-P-17667 or MIL-P-130.

5.2.1.2 Alternate unit protection. - Five fittings of one type and size, up to and including 1-1/2 inches in any dimension, may be packaged in perforated tear-apart cartons in accordance with Method III of MIL-P-116. Each carton shall have interior separations capable of supporting and separating each



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individual fitting in such manner as to prevent movement or contact of the fittings with each other. All surfaces of the carton coming in direct contact with the item shall be coated with a neutral barrier capable of withstanding slight abrasion. Fittings over 1-1/2 inches shall be packaged individually in accordance with 5.2.1.1.

5.2.1.3 Nuts, sleeves, and caps, unit containers. - Nuts, sleeves, and cap of one type and size shall be unit packaged in accordance with Method IC-1 of MIL-P-116, or in friction-top hermetically sealed cans conforming to PPP-C-96. The quantity per unit container shall be as specified in table X.

TABLE X. Quantity per unit container

| Size (inches) | Number |
|---|---|
| 1/8 to 1/2 | 100 |
| 5/8 to 1-1/4 | 25 |
| 1-1/2 to 3 | 5 |
| over 3 | 1 |

5.2.1.4 Fitting intermediate containers. - Unit packages shall be packaged in intermediate containers conforming to PPP-B-636, PPP-B-566, PPP-B-676, or PPP-B-665. The quantity of unit packages per intermediate container shall be as specified in table XI, except that intermediate containers are not required for fittings individually packed in unit cartons in accordance with 5.2.1.1.

TABLE XI. Quantity of packages

| Size (inches) | Quantity of unit packages per intermediate container | Quantity of tear-apart cartons per intermediate container |
|---|---|---|
| 1/8 through 1/4 | 100 | 20 |
| 5/16 through 1/2 | 25 | 10 |
| 5/8 through 1 | 10 | 5 |
| 1-1/4 through 3-1/2 | 5 | 1 |

5.2.2 Level C. - All fittings, nuts, sleeves, and caps shall be packaged as necessary to prevent damage or deterioration during shipment.

5.3 Packing. - Packing shall be level A, B, or C, as specified (see 6.2).



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5.3.1 Level A. - Fittings, packaged as specified herein, shall be packed in overseas-type shipping containers conforming to PPP-B-601, PPP-B-621, PPP-B-585, PPP-B-591, or PPP-B-636. Shipping containers, insofar as possible, shall contain the same number of articles, shall be uniform in shape and size and snugly packed, shall be of minimum cube and tare consistent with the protection required, and shall contain identical quantities. The gross weight of the fully packed exterior shipping container shall not exceed approximately 200 pounds, except fiberboard containers, which shall not exceed the weight limitations of the applicable container specification. Plywood, when used, shall be in accordance with NN-P-530, as applicable. Strapping and closures shall be in accordance with the appendix to the applicable container specification. Containers conforming to PPP-B-585 or PPP-B-621 shall be provided with a case liner conforming to MIL-L-10547, except that material shall conform to MIL-B-13239, and shall be sealed in accordance with the appendix to MIL-L-10547. No case liner is required when material conforming to type SF, class weather-resistant of PPP-B-636 is used as an intermediate container.

5.3.2 Level B. - Fittings packaged as specified herein shall be packed in domestic-type shipping containers conforming to PPP-B-636, PPP-B-585, PPP-B-591, PPP-B-601, or PPP-B-621. Closures shall be in accordance with the appendix to the applicable container specification. Fiberboard, when used, shall have a minimum Mullen test of not less than 275 pounds. The gross weight of the shipping container, when packed for shipment, shall not exceed 200 pounds; fiberboard containers shall not exceed the weight limitations specified in the applicable container specification.

5.3.3 Level C. - Packages which require overpacking for acceptance by the carrier shall be packed in commercial exterior shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.4 Marking of shipments. - If tear-apart containers are used, each section of the unit container shall be marked with the part number of the item. In addition to any other special marking required by the contract or order, marking for shipment shall be in accordance with MIL-STD-129.

## 6. NOTES

6.1 Intended use. - Fittings covered by this specification are intended for use in aeronautical fluid systems (see 1.1).

6.1.1 Since fittings of several materials are described in this specification, and various tube-fitting material combinations can be selected for various systems, table XII and the following information is offered:

(a) Under some circumstances, the fittings above may be used successfully at temperature and pressure ranges beyond those listed. Prior to such use, supporting test data should be approved by the procuring activity.



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(b) When corrosion-resistant steel sleeves are used with corrosion-resistant steel tubing, the tubing should be of the annealed type conforming to MIL-T-8504 or equal.
(c) Use of a single material by type for a stainless steel fitting assembly combination is not a recommended practice, as freezing or galling of threads with mating parts may result, and removal or disassembly cannot be accomplished without injury to the parts affected.
(d) Qualification approval of the product specified herein approves the product only for use in such manner as defined in the qualification tests.

TABLE XII. Pressure and temperature rating for fittings.

| Tube OD (nom) size | Aluminum alloy | | Carbon steel | | Corrosion-resistant steel | |
|---|---|---|---|---|---|---|
| | Pressure | Temperature | Pressure | Temperature | Pressure | Temperature |
| -2 through -12 incl. | 3,000 psi | -65° to +275° F | 3,000 psi | -65° to +275° F | 3,000 psi | -65° to +275° F |
| -16 through -32 incl. | 1,500 psi | | 1,500 1/ psi | | 1,500 1/ psi | |

1/ Carbon steel and corrosion-resistant steel fittings of -16 size may be used for 3,000 psi fluid systems.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Part No. required (see 1.2 and 3.3).
(c) Applicable levels of preservation, packaging, and packing (see 5.2 and 5.3).

6.3 Qualification. - With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Navy Department, Washington, D.C. 20360. However, information pertaining to qualification of products may be obtained from the Naval Air Development Center, AMD, Code MAEF, Johnsville, Warminster, Pennsylvania 18974.



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6.4 International standardization agreement. - Certain provisions of this specification are the subject of international standardization agreements ASCC AIR STANDARDS 17/15 and 17/18. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

6.5 Marginal indicia. - Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AV
Navy - AS
Air Force - 82

Reviewer activities:
Army - AV, WC, MI
Navy - AS
Air Force - 11, 82

User activities:
Navy - OS
Army - GL, EL, ME, AT

International interest (see 6.4)

Preparing activity:
Navy - AS

Project No. 4730-N0748



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